July 2023 CSE 102
Online 1 (Set 1)
Conditional Statement & Loops (by RRM)

Full Marks: 10

Finding Largest and Lowest Armstrong Numbers

Problem Statement:

Take two positive integers (*m, n*) of 32 bits as inputs that indicate the "lower limit" and "upper limit" respectively ($m<n$) of a range of positive integers. Find the largest and lowest "**Armstrong numbers**" in the given range. If there is no "Armstrong Number" between the lower and upper limit, print "-1".
N.B. An **Armstrong number** is a number that is the sum of its own digits; each raised to the power of the number of digits.

Example: 153 is an Armstrong number as $1^3 + 5^3 + 3^3 = 153$

Input: Two positive integers *m* and *n* (32 bits positive integers)

Output: Print the largest and lowest "**Armstrong numbers**".

Restrictions:

You cannot use one/multi-dimensional arrays or any other data structures. You cannot call any library functions (except scanf, printf, and pow).

Sample Inputs/Outputs:

Input	Output
1 1000	407 1
100 400	371 153
400 1000	407 407
500 1000	-1

Submission Guidelines:

- ➔ Solve the problem and rename the file as <your_id>.c (Example: student with ID 2205999 will rename it as 2205999.c).
- ➔ Then upload the .c file in Moodle.
- ➔ **Deadline: January 13, 2024 (Saturday)**
- ➔ **Do not copy from your classmate. In this case both of you will get -100%. Also do not copy code from any website or ChatGPT. You will get -200% in this case.**